

Do colours influence the effects of food labelling?

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Introduction

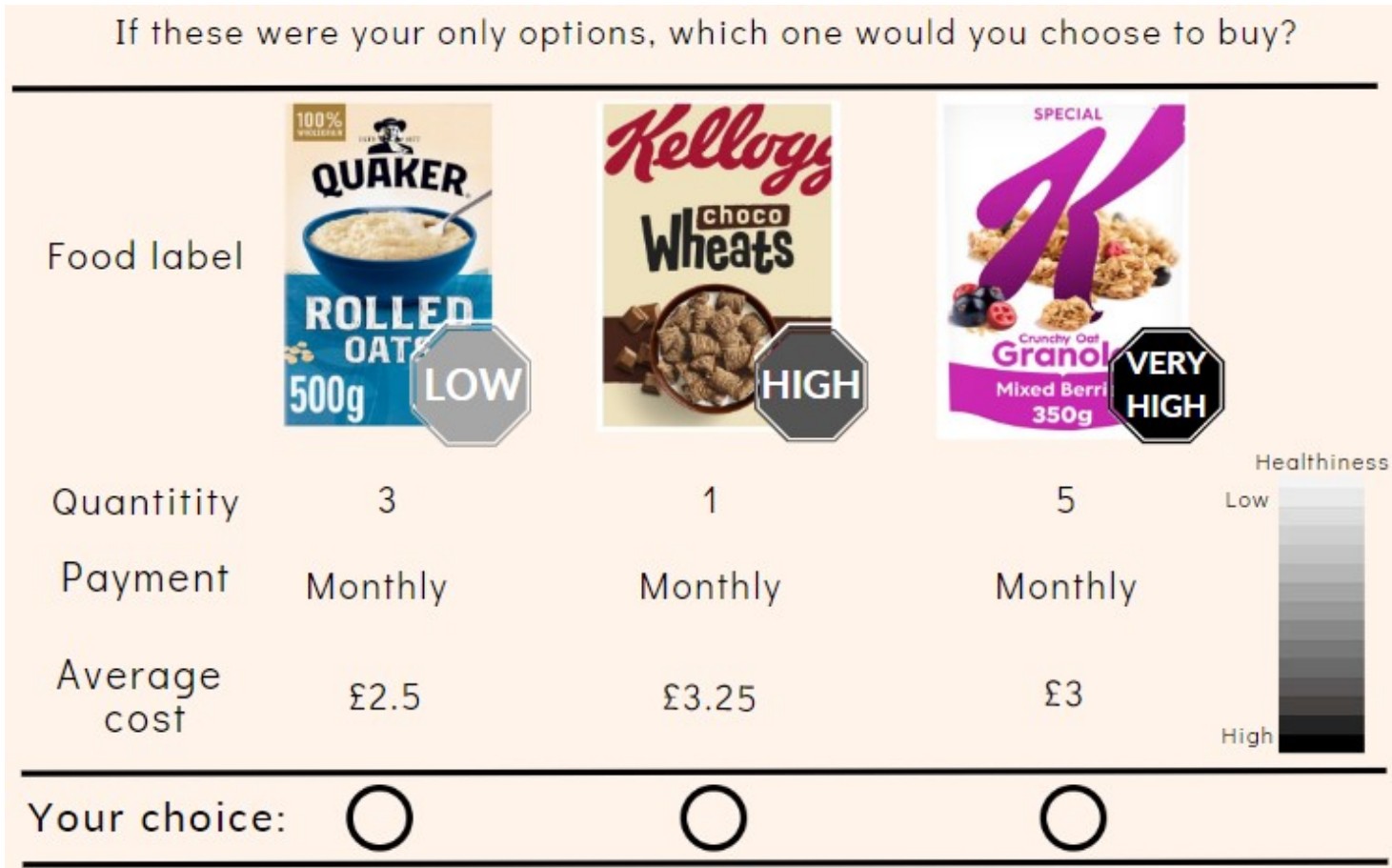
I propose to assess which design provides sufficient detail for the label to be interpreted optimally and translated into healthier choices. My hypothesis is that Traffic light labels (TL), as compared to grayscale labels (GS), increase the choice share of more nutritionally healthier cereals.

Objectives

1. Provides information to the public on what is the best label design to guide consumers’ choices.

Methods

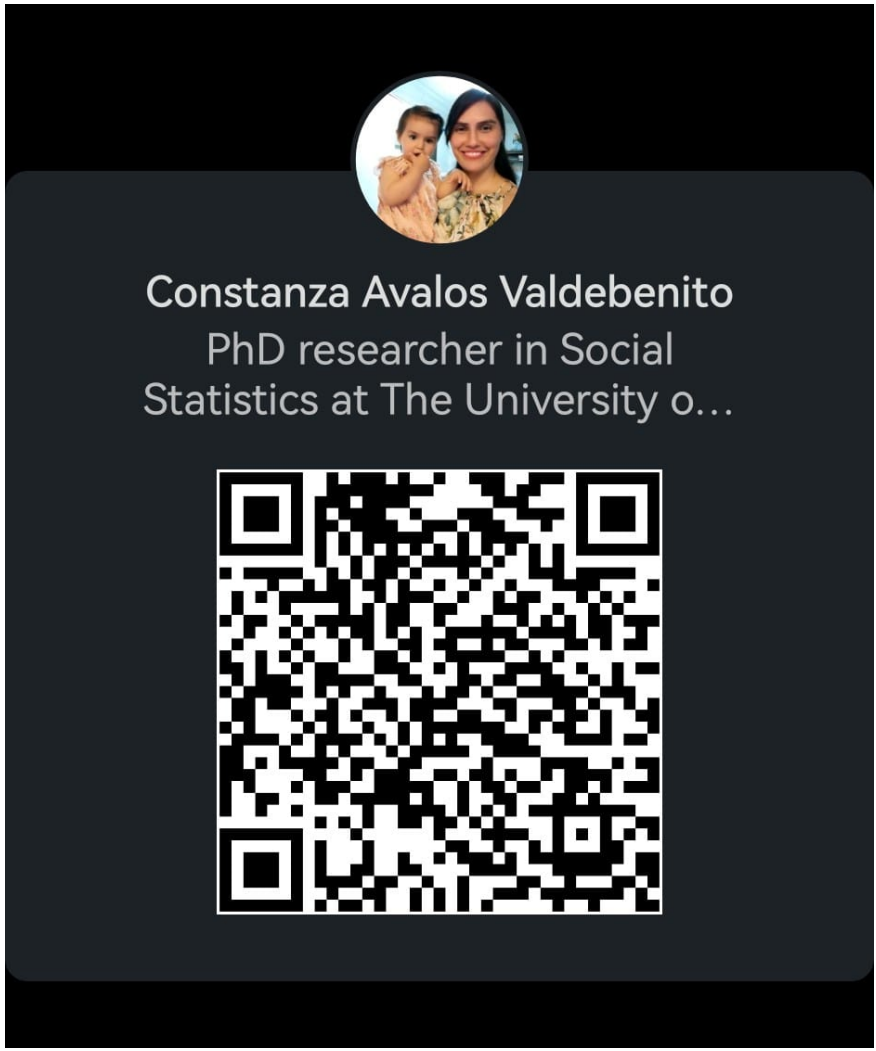
This simulated data is based on a conjoint analysis of 2,000 participants. In both the TL label and the GS label treatment group, participants were presented with 5 choice tasks, each of which included 3 products with different levels of calories.



Results

In a between-subjects design, they were randomly allocated to either the TL condition or the grayscale

Despite the information available, do consumers interpret food labels accurately?



condition. The distribution of controls among these groups did not significantly differ, as shown in Table 1.

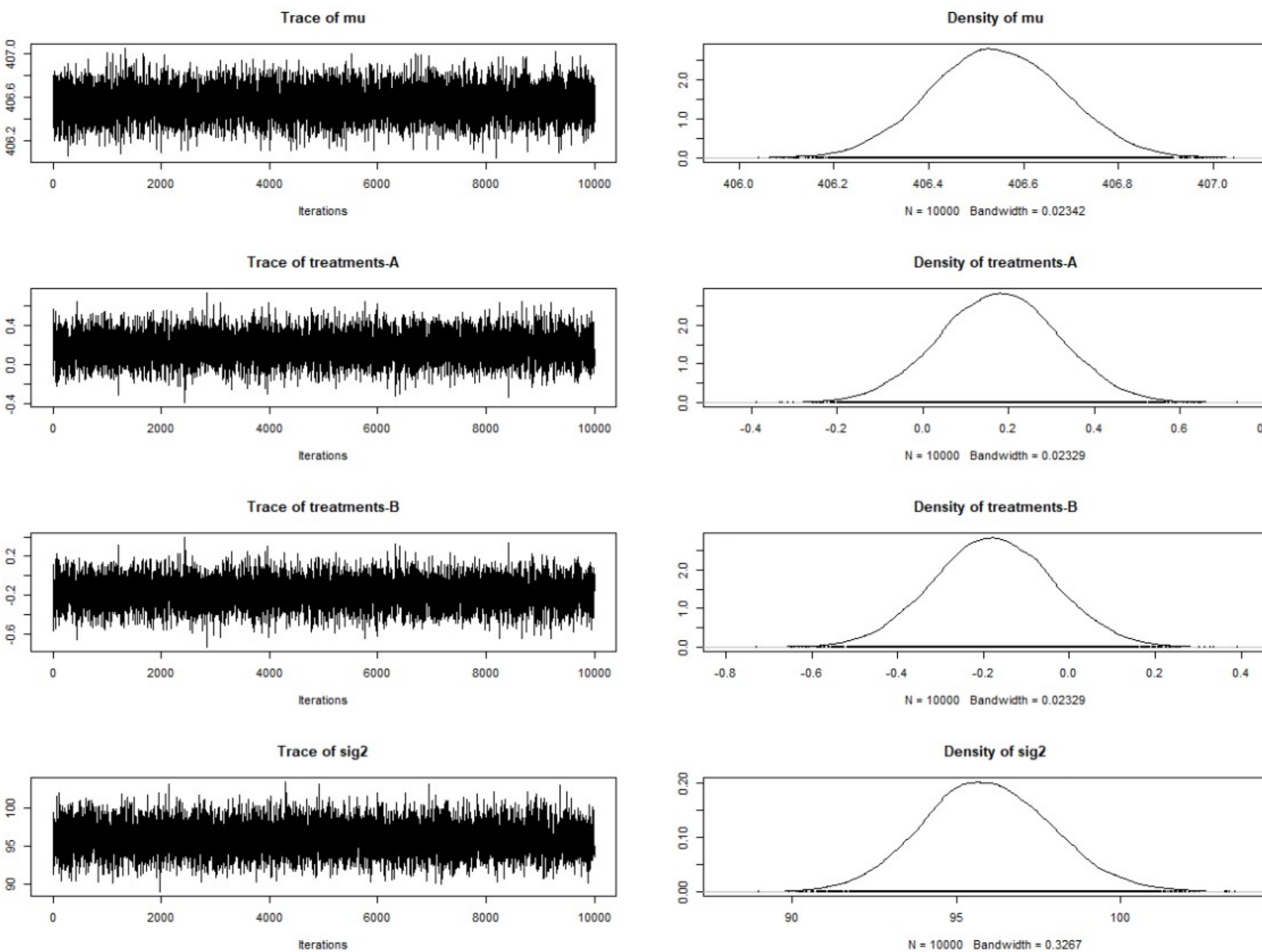
Table 1: Sample characteristics.			
Characteristics	Sample TL (n=1000)	Sample GS (n=1000)	Test statistics
Age 35-44, n (%)	146 (14.5%)	145 (14.6%)	-
Female, n (%)	335 (33.5%)	334 (33.5%)	-
Raven score, mean (2-sided)	30.02	31.89	t=-16.17 (p=1.55)

Table 2 shows the coefficients of the mixed logit model used to estimate the effect of labels on cereal simulation choices by TL and GS condition. Comparing the high choices, the coefficient of high/black dark (-0.758) is higher than that of high/red (-0.710). For medium choices, medium/yellow has higher coefficients (-0.805) than medium/light dark (-0.793).

Table 2. Estimation results of Model 1 (TL) and Model 2 (GS) —predicting product choice

	Dependent variable	
	choice1	choice2
	(1)	(2)
cereal_labelMediumTL	-0.805 ^{***} (0.047)	
cereal_labelHighTL	-0.710 ^{***} (0.047)	
cereal_labelMediumGS		-0.793 ^{***} (0.047)
cereal_labelHighGS		-0.758 ^{***} (0.047)
quantity3	-0.422 ^{***} (0.045)	-0.519 ^{***} (0.046)
quantity5	-1.026 ^{***} (0.049)	-1.055 ^{***} (0.049)
price3	-0.823 ^{***} (0.045)	-0.883 ^{***} (0.046)
price3.25	-1.599 ^{***} (0.051)	-1.620 ^{***} (0.051)
Observations	5,000	5,000
Log Likelihood	-4,554.915	-4,524.288
Note:	p<0.1; p<0.05 ; p<0.01	

What would happen if we use the Bayes factor approach? If we compare a regression **m1 <- lmBF(mean_calories ~ treatments vs m2 <- lmBF(mean_calories ~ treatments + new_col_raven, data = df)**, it would give us plots with a lower bandwidth resulting in a more detailed and irregular density estimate by conditions. We add more uncertainty to the prediction.



References

Ravaioli, S. (2021). Coarse and Precise Information in Food Labeling. Job Market Paper, 1.